1.41 Graph Sketching: Trigonometric Functions in Degrees

Worked Solutions

1 a)

$y = 2\sin x$.

1 b)

$y = 4\sin x$.

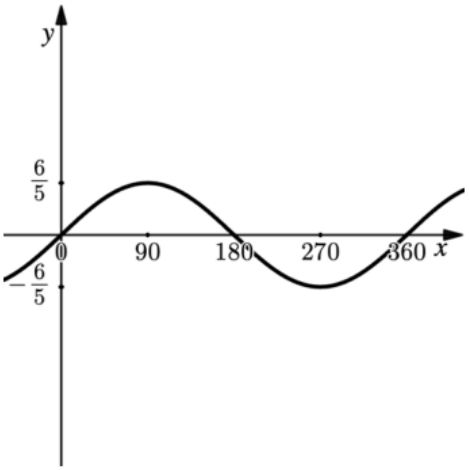
2 a)

$y = \frac{1}{2}\cos x$.

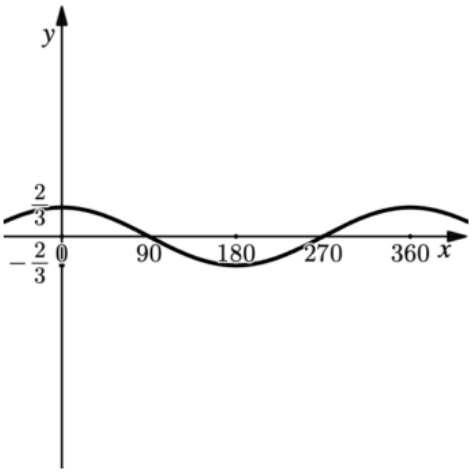
2 b)

$y = \frac{3}{2}\cos x$.

3 a)



3 b)



4 a)

$y = \sin 2x$.

4 b)

$y = \sin 3x$.

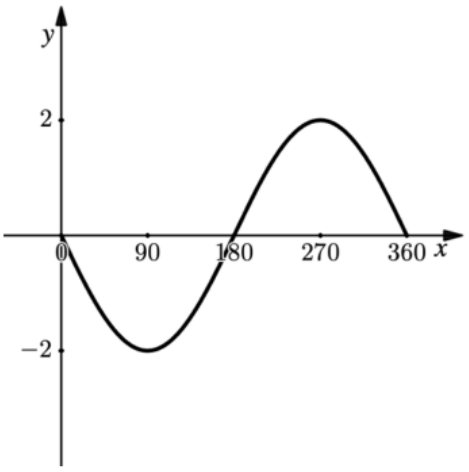
4 c)

$y = \sin \frac{3}{2}x$.

4 d)

$y = \sin \frac{1}{2}x$.

5)



6 a)

$y = \cos 3x$.

6 b)

$y = \cos 4x$.

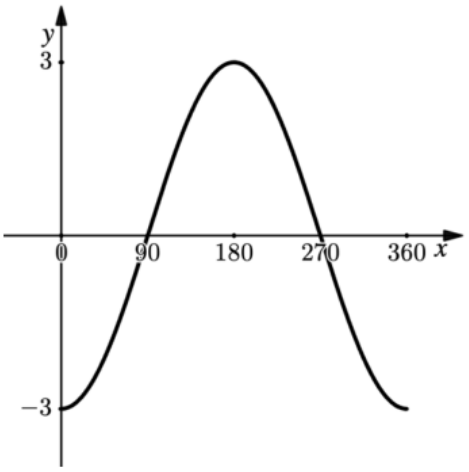
6 c)

$y = \cos \frac{4}{3}x$.

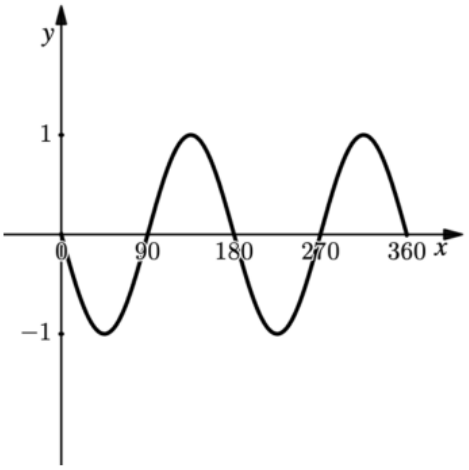
6 d)

$y = \cos \frac{1}{3}x$.

7)



8)



9 a)

$y = \tan 2x$.

9 b)

$y = \tan \frac{1}{2}x$.

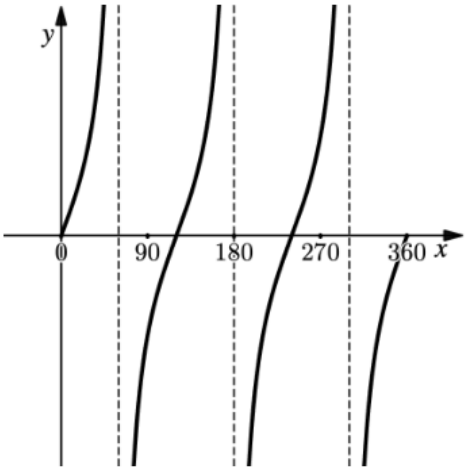
9 c)

$y = \tan \frac{1}{3}x$.

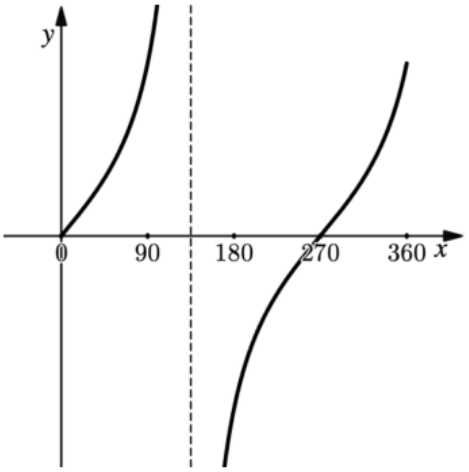
9 d)

$y = \tan \frac{5}{3}x$.

10 a)



10 b)



11 a)

$y = 2 + \sin x$.

11 b)

$y = 1 + \cos x$.

11 c)

$y = -1 + \sin x$.

11 d)

$y = -3 + \cos x.$

12 a)

$y = 1 + 2\sin x.$

12 b)

$y = 2 + 3\cos x.$

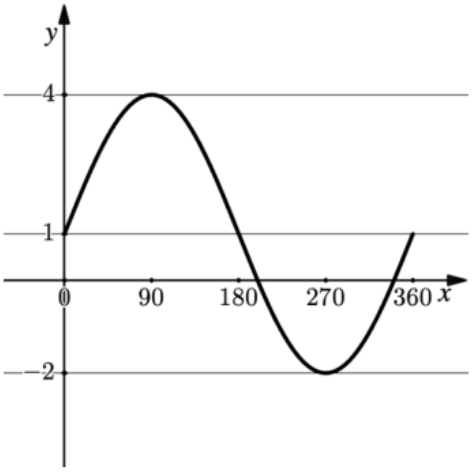
12 c)

$y = -1 + 3\sin x.$

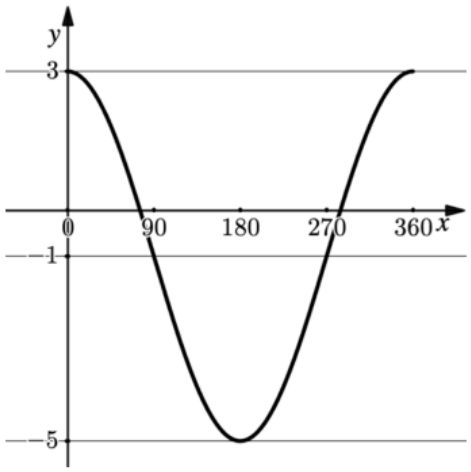
12 d)

$y = -2 + 2\cos x.$

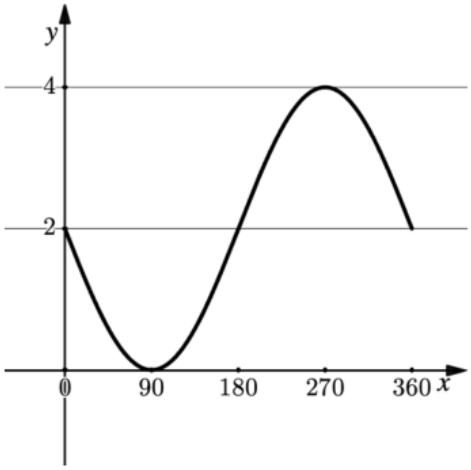
13 a)



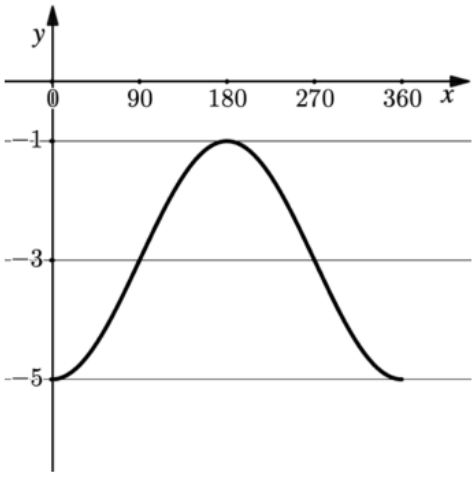
13 b)


y
3
0
90
180
270
360
x
−1
−5

13 c)


y
4
2
0
90
180
270
360
x

13 d)


y
0
90
180
270
360
x
−1
−3
−5

14 a)

$y = \sin(x + 30°)$.

14 b)

$y = \sin(x - 60°)$.

15 a)

$y = \cos(x - 60°)$.

15 b)

$y = \cos(x - 150°)$.

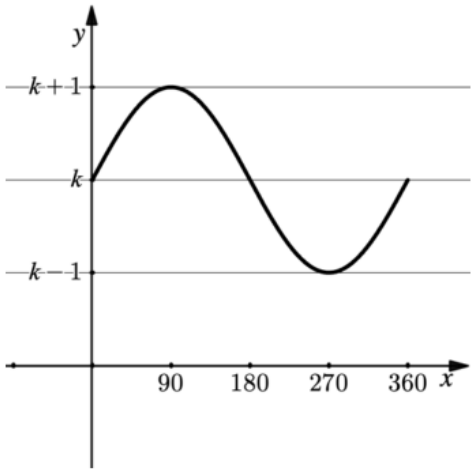
16 a)

$y = \tan(x + 30°)$.

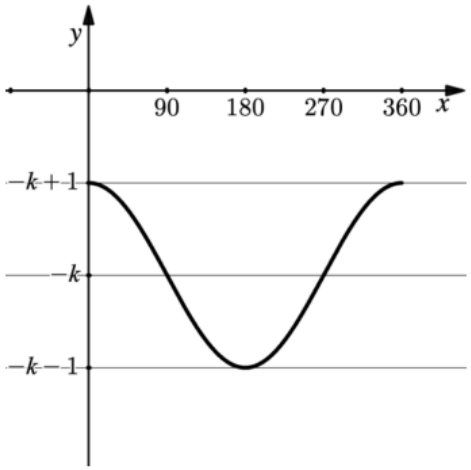
16 b)

$y = \tan(x - 45°)$.

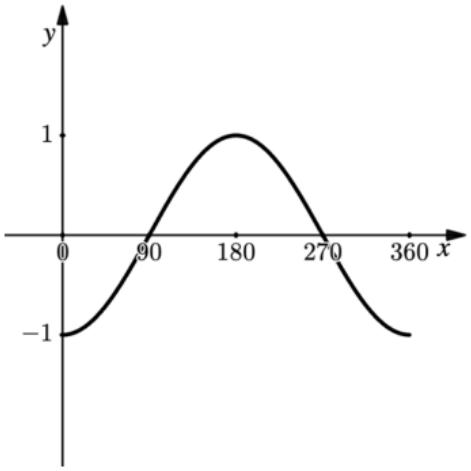
17 a)



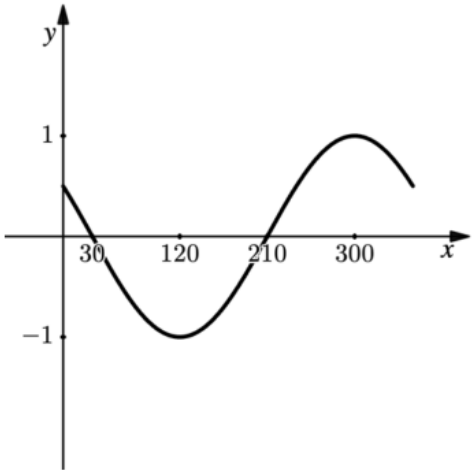
17 b)



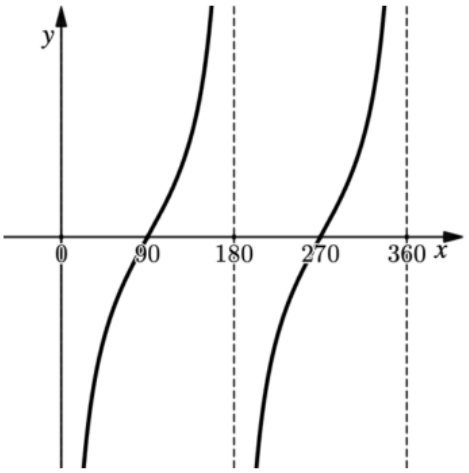
18 a)



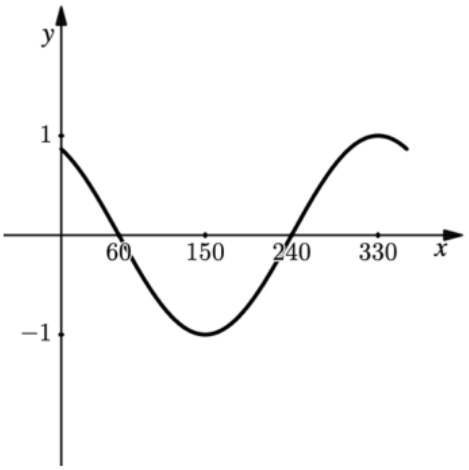
18 b)



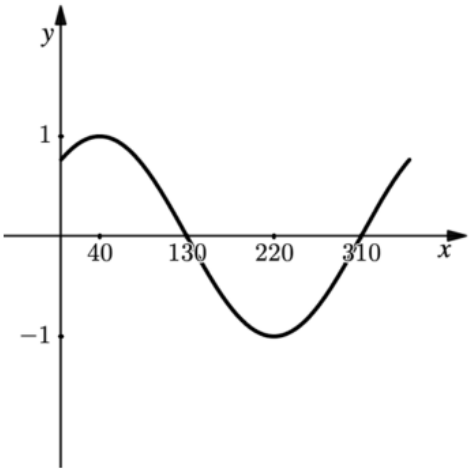
18 c)



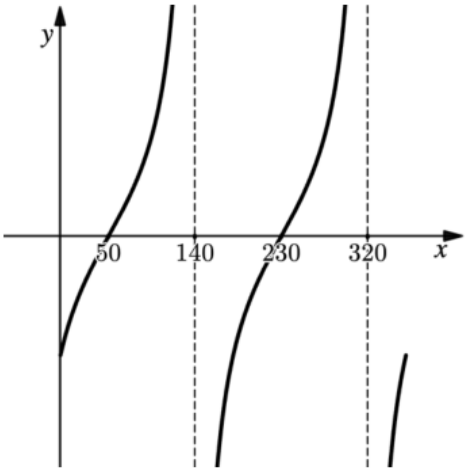
18 d)



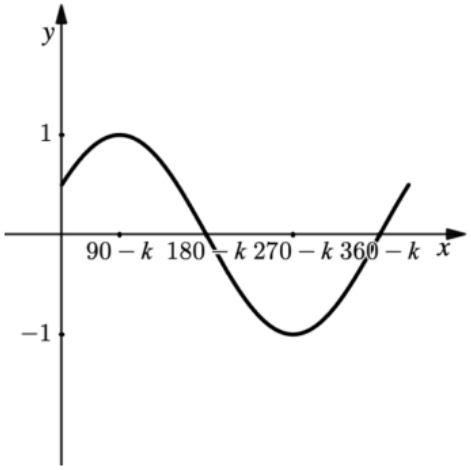
18 e)



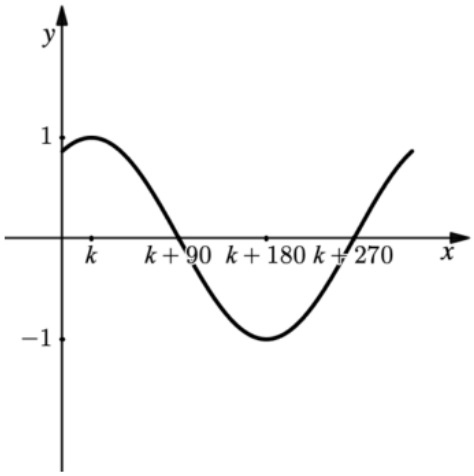
18 f)



19 a)



19 b)



20 a)

$y = \sin(2x + 90°)$.

20 b)

$y = \sin(\frac{1}{2}x - 30°)$.

20 c)

$y = \sin(\frac{3}{2}x + 60°)$.

21 a)

$y = \cos(3x - 90°)$.

21 b)

$y = \cos(\frac{3}{4}x + 66°)$.

21 c)

$y = \cos(\frac{5}{2}x - 100°)$.

22 a)

$y = \tan(2x + 120°)$.

22 b)

$y = \tan(\frac{1}{2}x - 60°)$.

22 c)

$y = \tan(\frac{2}{3}x + 36°)$.

23 a)

$y = 1 + \sin(\frac{1}{2}x + 40°)$.

23 b)

$y = -1 + 2\sin(x - 80°)$.

23 c)

$y = -2 + 2\sin(2x - 30°)$.

23 d)

$y = 1 + 3\sin(\frac{3}{2}x - 96°)$.

24 a)

$y = 1 + 2\cos(x - 60°)$.

24 b)

$y = -1 + 3\cos 2x$.

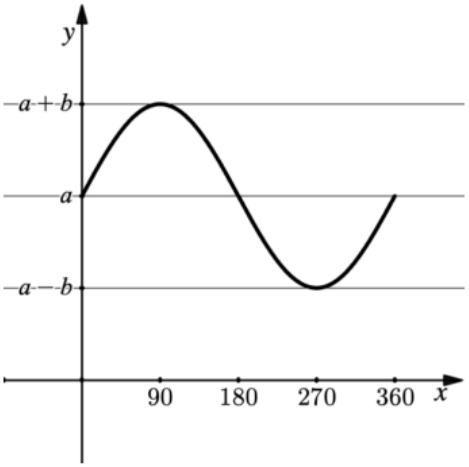
24 c)

$y = -2 + \cos(\frac{6}{5}x + 30°)$.

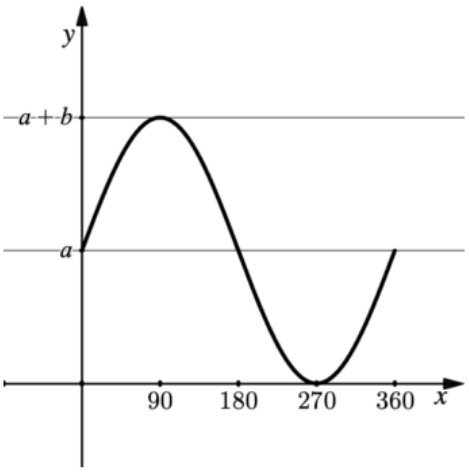
24 d)

$y = 2 + 4\cos(\frac{3}{5}x - 120°)$.

25 a)



25 b)



25 c)

